

Homework Instructions

Homework 2 can be completed either individually or in groups of two. Please indicate clearly in your submission whether the homework is submitted individually or by a group of two.

Homework 2 revolves around the use of the U.S. Economic Policy Uncertainty indicator in combination with a VAR.

When you submit your homework answers, please include the written answers and plots in a single PDF. In addition, please submit a separate folder containing the code and data.

Whenever you are asked to plot the data, please include the plot in the PDF that you submit. The figures must be clearly labelled and must contain a figure note describing what the reader sees in the figure. In other words, the figure and the figure note should be sufficient for any reader to understand what is shown in the figure and how the results were produced. This will be part of the evaluation of Homework 2.

- **Deadline:** 19 June 2026 at 10:00am

Part I

1. Construct the dataset. The file `DataFigure.png` shows how the series should look if you downloaded them correctly.
 - a. Download the U.S. Monthly EPU Index from policyuncertainty.com.
 - b. Download the monthly non-farm employment, personal consumption expenditures price index, and industrial production series from <https://fred.stlouisfed.org/>.
 - i. Download the series in levels and seasonally adjusted.
 - ii. The mnemonics to find the series are `PAYEMS`, `PCEPI`, and `INDPRO`.
 - iii. Download the series from 1985M1 to 2023M3.
2. Estimate a VAR with one lag using the following ordering of variables:

`News_Based_Policy_Uncert_Index`, `PAYEMS`, `INDPRO`, `PCEPI`.

The first variable is the Economic Policy Uncertainty index.

- a. The variables `PAYEMS`, `INDPRO`, and `PCEPI` enter the model in month-on-month growth rates, i.e.,

$$100 \times \left(\frac{X_t}{X_{t-1}} - 1 \right).$$

- b. The EPU index enters the model in levels.
- c. Estimate the model using only data from 1985M2 to 2019M12.
- d. Report the estimated reduced-form VAR(1). Write down the four estimated equations, including the estimated constant and lag coefficients.
- e. Report the estimated reduced-form variance-covariance matrix of the residuals.

3. Using the same VAR, estimate impulse response functions for a shock to the Economic Policy Uncertainty index.
 - a. Plot the impulse response functions for the month-on-month growth rates of employment and industrial production, including 95% confidence bands.
 - b. Include the plots in your answer PDF and describe what you see. How do employment growth and industrial production growth react to an economic policy uncertainty shock?

Part II

Imagine you work in a central bank or a consulting firm and you are asked to provide forecasts for U.S. industrial production and employment.

1. Using the same model as in Part I, estimated on data from 1985 to 2019, produce unconditional forecasts for 12 horizons ahead using March 2023 as the latest data point.
 - a. Plot the forecasts for month-on-month employment growth and month-on-month industrial production growth.
 - b. Briefly describe the forecast trajectory of both variables. In particular, discuss whether the forecasts predict positive or negative growth, and whether growth increases or decreases over the forecast horizon.
2. Imagine there were national U.S. elections in May 2023. Produce a structural forecast, assuming that the paths of the U.S. EPU index in April, May, June, July, and August 2023 are the same as in October 2020, November 2020, December 2020, January 2021, and February 2021, respectively. These months correspond to the period before, during, and after the last U.S. national elections.
 - a. Use the ordering of the variables described above, i.e.,

`NewsBased_Policy_Uncert_Index, PAYEMS, INDPRO, PCEPI.`

 The EPU index is ordered first.
 - b. For all variables other than the EPU index, the last observed value should be March 2023.
 - c. Plot the forecasts for industrial production and employment.
3. Are the forecasts from parts (1) and (2) different? If so, why? What does the structural forecast mean? Please provide a concise answer.
4. Point out three shortcomings of the analysis conducted in parts (1) to (3). Present them as bullet points. The shortcomings can be related to any aspect of the exercise, including the model type, the construction of the EPU index, the identification assumptions, or the forecasting strategy.